SING's Cross-Model Leakage Ranking Is Indistinguishable from a Null-Energy Ranking

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Abstract

SING (Yadid et al., 2026) ranks 13 ImageNet classifiers by how much class semantics "leak" into the null space of their linear heads, measured by the angular shift (AS) of a CLIP-aligned translation when the null component is removed. We audit this ranking. The fraction of feature energy in the null space, ρ, reproduces SING's published order as well as SING's own score does (Kendall τ=0.82 for both; paired difference 95% CI $[-0.36, 0.29]$), and each keeps a partial association with the order given the other (τ=0.61 both ways). On this evidence, SING's ranking is indistinguishable from a null-energy ranking. Interventions point to energy as a confound in a specific sense. With SING's released translators, removing the same fraction of energy from every model's null space removes the agreement ($\tau = -0.03$ and 0.09 on two model sets), and so does matching the removed energy in CLIP space. Replacing each model's null content with image-independent Gaussian noise of the same mean and covariance leaves the logits unchanged, yet AS still grows with the injected energy on all three models tested and the resulting scores reproduce SING's order (τ=0.69). AS therefore has a path from null energy to the score that carries no image-specific content. Retraining the translators preserves the agreement (τ=0.85), although the equal-energy result depends on the translator. With retrained ridge translators, equal-energy AS keeps part of the agreement (τ=0.61 and 0.75). We also find that SING's "DinoViT" translator matches a supervised timm ViT-L/16 checkpoint, with the other public candidates at chance. Size-corrected scores give orders that differ from SING's, but none reverses it with confidence. All analyses were pre-registered over three rounds, and we report every outcome, including two killed score searches. We recommend reporting ρ and an energy-only baseline next to any cross-model null-space score.

1 Introduction

A linear classifier W on features $f \in \mathbb{R}^m$ ignores every component of f in its null space $\ker W$. SING (Yadid et al., 2026) reads class-relevant content in that null space as "leakage": information the head could have used but did not. To measure it, SING maps features into CLIP space (Radford et al., 2021) with a learned linear translator, removes the null component, and records how far the translated feature turns relative to the class text (the absolute shift, AS). It uses AS and an IS/AS ratio to rank 13 ImageNet models and reports that a model labelled "DinoViT" (Caron et al., 2021) leaks least.

Null spaces matter well beyond SING. They appear in work on out-of-distribution failure (Idnani et al., 2023), outlier and OOD detection (Wang et al., 2022; Cook et al., 2020; Zöngür et al., 2025), overfitting (Rezaei & Sabokrou, 2023) and information hiding (Li & Short, 2024). Invariance-based adversarial examples show that class information can sit in directions a classifier ignores (Jacobsen et al., 2019). In NLP, nullspace projection and closed-form erasure remove a property from representations (Ravfogel et al., 2020; Elazar et al., 2021; Belrose et al., 2023). Linear maps between a vision model and CLIP, the kind of translator SING relies on, have been studied by Moayeri et al. (2023). Probing work has also shown that readouts need controls (Hewitt & Liang, 2019), and similarity scores are known to be dominated by high-variance directions (Davari et al., 2023).

Across architectures, the null dimension d_{null} and the null energy fraction

$$\rho = \mathbb{E}\|\Pi f\|^2 / \mathbb{E}\|f\|^2, \qquad \Pi = V_n V_n^\top,$$

vary by orders of magnitude (Table 5). Removing more energy should move a translated feature further whether or not that energy is semantic. If so, a cross-model comparison of null "semantics" would be tied to energy. This paper tests that possibility.

Contributions.

1. **Energy-only control** (§3). ρ alone reproduces SING's 13-model order as well as SING's score. The two are statistically indistinguishable predictors, and neither subsumes the other.
2. **Interventions** (§4). We test the role of energy with equal-energy removal, matched CLIP-space energy, leak-free surrogate null content and retrained translators. With the released translators, energy accounts for the agreement. With retrained translators, it accounts for part of it.
3. **The DinoViT checkpoint** (§5). SING's translator for this model fits a supervised timm ViT-L/16; the other public candidates are at chance.
4. **Size-corrected scores** (§6). They change the order, but no correction reverses it with confidence. Several class-level claims do not reproduce on our pipeline.
5. **Full reporting**. Three rounds of pre-registration with every outcome listed (Appendix A), including the kills of two score searches and the flaw in the flatness rule behind them (Appendix D).

2 Setup

SING's score. Let $T(f) = Af + b$ be SING's released linear translator into CLIP space, W the model's head and V_n an orthonormal basis of $\ker W$ from the SVD of W (absolute singular-value tolerance 10^{-5} unless stated). For a text direction t, the per-image absolute shift for a removed component r is $|\theta(T(f), t) - \theta(T(f - r), t)|$. AS_{SING} is its mean over images with $r = \Pi f$ (full null removal) and the class text (the normalised mean of SING's six prompts) as t. We use SING's released code (commit `e47b37d`) and all 13 released translators on a fixed 10k-image ImageNet-val subset (10 images per class, all 1000 classes). SING's own model comparison also uses 10k features over the 1000 classes; its 16-class, 2000-image subset is the translator-validation set, which we do not use. For the class-level claims we add 200 images (all 50 validation images of jellyfish, porcupine, Arabian camel and sports car).

Reference order and statistics. SING's order is the 13 IS/AS bars of its Supp. Fig. 12b, which we digitised. A lower ratio means more leakage. Agreement is Kendall τ_b between a leakage score and that order. Model-level uncertainty comes from 2000 bootstrap resamples of models (seed 0), with percentile and BCa intervals (Efron, 1987). Paired comparisons of two scores use the same resamples. With n=13 these intervals are wide and often reach 1, so we read them as coarse. Sets: *all 13*; *E12*, the 12 models with $\rho \geq 0.005$ (all but DenseNet121, whose ρ=1.8×10^{-8}); and *S11*, a screened set declared in round 2 that also drops BiT. The screen removes models with $\rho < 10^{-4}$ or with a retained row-space direction of singular value $\sigma < 10^{-3}$ that holds more than 5% of the feature energy. BiT has such a direction (σ=8.9×10^{-4}, 26% of the energy). ResNet50 has one just under the threshold ($\sigma < 10^{-3}$, 4.4% of the energy): at tolerance 10^{-4} that direction joins the null space, ρ rises from 0.146 to 0.190 and AS_{SING} falls from 1.68° to 1.33°.

Pre-registration. Every experiment below was declared in writing before its numbers were computed, over three rounds. The ledger is in Appendix A. Results marked *post hoc* were added after seeing data.

3 Null energy reproduces SING's order

Reproduction. AS_{SING} agrees with SING's order at τ=0.82 (95% CI $[0.56, 1.00]$). The like-for-like comparison, our IS/|AS| ratio against SING's IS/AS ratio, gives τ=0.744 (percentile $[0.33, 1.00]$, BCa $[0.12, 0.94]$).

Model	ρ	ours IS/|AS|	SING IS/AS	SING/ours
VGG16	0.350	17.75	24.26	1.37
VGG19	0.339	15.56	19.89	1.28
DenseNet121	1.8×10^{-8}	41.48	70.71	1.70
ResNet50	0.146	17.40	24.81	1.43
"DinoViT"	0.005	20.95	70.16	3.35
EffNet-B0	0.130	22.15	36.28	1.64
BiT-R50x1	0.046	24.93	47.21	1.89
ResNeXt101	0.451	12.60	13.88	1.10
ConvNeXt-B	0.013	32.86	37.38	1.14
Swin-L	0.274	21.21	29.18	1.38
DINOv2-L	0.322	15.49	16.61	1.07
DeiT3-L	0.348	9.73	9.51	0.98
EVA02-L	0.009	21.64	54.32	2.51

Table 1: Like-for-like reproduction. Our IS/|AS| on the 10k subset against SING's IS/AS (digitised Supp. Fig. 12b). τ(ours, SING)=0.744 [0.33, 1.00].

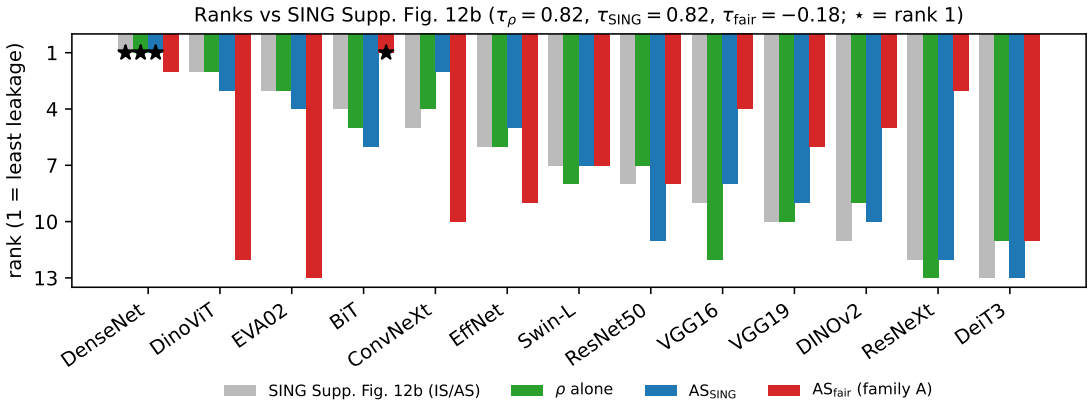


Figure 1: Ranks (1 = least leakage, ⋆ marks rank 1) under SING Supp. Fig. 12b, ρ alone, AS_{SING} and the size-corrected AS_{fair} (family A, §6).

SING's ratios are 0.98–3.35 times ours (Table 1). The gap is largest for DinoViT (3.35×) and EVA02 (2.5×), both small-ρ models. SING's model comparison uses the same number of images and classes as ours, so the gap is not explained by the subset size. Its likely sources are SING's specific images, its concept set or its checkpoint (§5), and we cannot separate them without SING's evaluation lists.

ρ alone reproduces the order. Ranking models by ρ alone gives τ=0.82 [0.56, 1.00], the same as AS_{SING} (Fig. 1). On S11 the values are 0.78 for ρ and 0.82 for AS_{SING}. The paired bootstrap difference $\tau(\rho) - \tau(AS_{SING})$ is 0.00 with 95% CI [−0.36, 0.29] (BCa [−0.37, 0.28]). Both agreements are far from chance (permutation p=5×10^{-5} for each, the smallest value 20000 permutations can give). ρ and AS_{SING} agree with each other at τ=0.69 (Spearman 0.87). Each keeps a partial association with SING's order given the other: partial Kendall τ=0.61 both ways, and partial Spearman 0.63 for AS_{SING} given ρ and 0.72 for ρ given AS_{SING}. So the data cannot tell the two predictors apart, and ρ does not make AS_{SING} redundant. Null dimension alone is a weak predictor (Spearman −0.46, p=0.12): DeiT3-L (Touvron et al., 2022) has d_{null}=24 yet the worst ratio. A NuSA-style null norm, $\mathbb{E}_i\|\Pi f_i\|/\|f_i\|$, in the spirit of Cook et al. (2020), is a second null-space score; it orders the models almost as ρ does (τ=0.97) and agrees with SING at τ=0.85.

Within a model, AS grows with removed energy. On ResNet50, ResNeXt101, BiT-R50x1 (Kolesnikov et al., 2020) and DINOv2-L (Oquab et al., 2024), all with d_{null}=1048, removing random d-dimensional subspaces of the null space with $d \in \{24, \ldots, 1048\}$ raises AS about tenfold from d=24 to d=1048 on three of four models (Table 9, Fig. 3 in Appendix C). For random subspaces, removed energy is close to linear in d, so this experiment cannot separate energy from dimension. A second experiment mixes high- and low-variance null PCs to vary energy at fixed d and dimension at fixed energy. It fits $\log|\text{AS}| = a\log\rho + b\log d + c_{\text{model}}$ over 76 cells on five models with a=0.431 (cell bootstrap $[0.388, 0.485]$, model-clustered bootstrap $[0.382, 0.464]$, leave-one-model-out 0.408–0.452; the clustered CI rests on five clusters, one with a single cell, so it is not more reliable for being narrower) and a dimension term b=0.010 whose CI covers 0 (Appendix C). A first-order calculation gives a=0.5: if the translator maps the removed component linearly and the residual is small, the angle changes in proportion to $\|\Pi f\| \propto \sqrt{\rho}$. The pre-registered flatness part of this experiment failed (max/min across d at fixed ρ up to 2.29 against a 1.5 bar), so we claim only $a \approx 0.4$–0.5.

BiT is the exception. Its full-null AS collapses (r=0.06) while its partial subspaces give large AS (9–13°), because the translator's null blocks nearly cancel (top singular value of AV_n 284, against 5.6 for ResNet50). This is also why BiT fails the screen.

4 Is energy a confound? Interventions

Agreement between ρ and SING's order does not show that energy drives AS across models. Energy could instead *mediate* genuine leakage, if models with more null energy also hold more class content there. We ran four interventions, all pre-registered in round 3, to separate the two readings (Table 2, Fig. 2).

4.1 Equal null energy (released translators)

Design. In *A1* we remove $\beta\Pi f$ with $\beta = \sqrt{\rho_0/\rho}$, ρ_0=0.005, so every model loses the same fraction of its feature energy, taken from its own null space in its own proportions. *A2* removes the same energy from the top-variance null PCs, and *A3* removes it from a random rotation of the null basis. *A4* matches energy in CLIP space instead: with $\kappa = \mathbb{E}\|A\Pi f\|^2/\mathbb{E}\|Tf\|^2$, we remove $\beta'\Pi f$ with $\beta'^2\kappa = \kappa_0$=0.0023. The pre-registered outcome was ENERGY-ACCOUNTS if $\tau(\text{A1}) < 0.3$ and the paired CI of $\tau(\text{AS}_{\text{SING}}) - \tau(\text{A1})$ is above 0 on both E12 and S11, and a kill of the energy reading if $\tau(\text{A1}) \geq 0.6$ on both.

Result. Equal-energy AS no longer agrees with SING: $\tau = -0.03$ $[-0.54, 0.61]$ on E12 and 0.09 $[-0.45, 0.78]$ on S11. The paired drop from AS_{SING} has CI $[0.23, 1.39]$ (BCa $[0.25, 1.41]$) on E12 and $[0.08, 1.30]$ on S11, so the outcome is ENERGY-ACCOUNTS. Taking the energy from top PCs (0.03) or random directions (0.03) gives the same picture, and so does matching CLIP-space energy (A4: -0.15 and -0.20). The A1 scores still differ across models by a factor of six (Table 5), so equalising energy changes their order without making them equal.

Translator gain. CLIP-space energy κ tracks SING's order slightly less well than ρ (τ=0.67 on E12). The paired difference $\tau(\kappa) - \tau(\rho)$ has CI $[-0.41, 0.28]$, so we keep ρ as the named variable. The translator's null gain $\gamma = \kappa/\rho$ does not track SING ($\tau = -0.06$). Differences in how strongly each translator amplifies null directions therefore do not explain the order.

4.2 Leak-free null content

Design. We replace each image's null coordinates $c_i = V_n^\top f_i$ with a draw $\tilde{c}_i \sim \mathcal{N}(\mu, \Sigma)$, where μ and Σ are the mean and covariance of c over the 10k images (5 draws). The row-space part, and so every logit, is unchanged. The surrogate carries no image-specific and thus no class-specific content. On ResNet50, ResNeXt101 and DINOv2-L we scale the surrogate to null energy fractions $\rho_t \in \{0.01, 0.05, 0.2, 0.45\}$. We then optionally plant a concept direction with strength $\alpha \in \{0, 0.1, 0.3\}$, using the generator of our holdout test (Appendix D; concepts water, green, thin, knitted). The pre-registered test (from the second review)

Score	all 13	E12	S11
ρ (null energy fraction)	0.82 [0.56, 1.00]	0.79 [0.49, 0.97]	0.78 [0.44, 1.00]
$\mathrm{AS}_{\mathrm{SING}}$ (released translators)	0.82 [0.56, 1.00]	0.79 [0.51, 1.00]	0.82 [0.51, 1.00]
κ (CLIP-space null energy)	0.72 [0.44, 0.95]	0.67 [0.37, 0.93]	0.67 [0.35, 0.96]
NuSA-style null norm	0.85 [0.64, 1.00]	0.82 [0.57, 0.97]	0.82 [0.53, 1.00]
$\gamma = \kappa/\rho$ (translator null gain)	−0.10 [−0.56, 0.43]	−0.06 [−0.59, 0.54]	0.05 [−0.51, 0.74]
Leak-free surrogate $\mathrm{AS}_{\mathrm{surr}}$	0.69 [0.39, 0.92]	–	0.64 [0.27, 0.92]
Equal-energy interventions (released translators)			
A1: remove $\beta\Pi f$, $\beta^2\rho = 0.005$	–	−0.03 [−0.54, 0.61]	0.09 [−0.45, 0.78]
A1′: same, $\rho_0 = 0.001$	–	−0.03 [−0.54, 0.61]	0.09 [−0.45, 0.78]
A2: top null PCs, energy ρ_0	–	0.03 [−0.54, 0.54]	0.13 [−0.43, 0.67]
A3: random null rotation, energy ρ_0	–	0.03 [−0.47, 0.67]	0.16 [−0.50, 0.84]
A4: matched CLIP-space energy κ_0	–	−0.15 [−0.68, 0.44]	−0.20 [−0.74, 0.46]
Retrained ridge translators (test half)			
$\mathrm{AS}_{\mathrm{SING}}$	0.85 [0.61, 1.00]	0.82 [0.57, 1.00]	0.82 [0.51, 1.00]
A1	–	0.61 [0.15, 0.94]	0.75 [0.32, 1.00]
A1, λ on extended grid†	0.49 [−0.10, 0.91]	0.39 [−0.21, 0.86]	0.53 [−0.06, 1.00]

Table 2: Kendall τ against SING's order (percentile 95% CI, 2000 model resamples). E12 drops DenseNet ($\rho < \rho_0$); S11 also drops BiT. A1′ gives the same order as A1 because, for each model, AS is monotone in the removed fraction. †Post hoc: ridge penalty chosen on an extended grid $10^{3..6}$, since the pre-registered grid put every model at its upper edge.

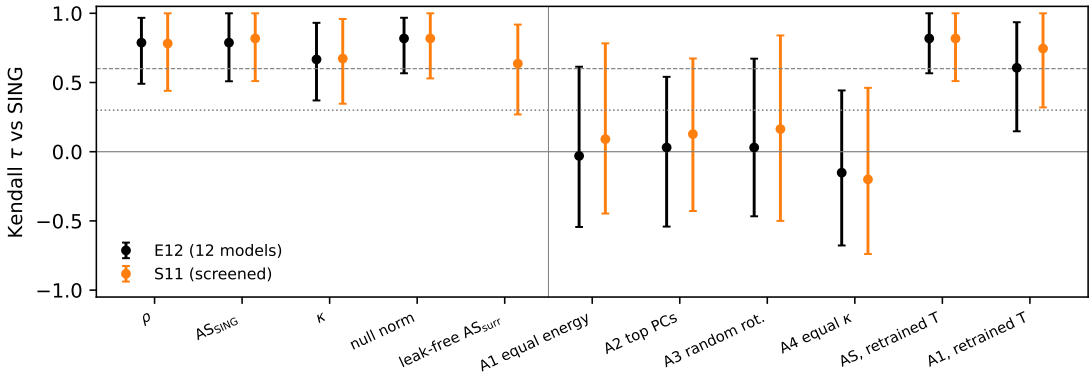


Figure 2: Agreement with SING's order under the energy controls and interventions of §4 (95% percentile CIs). Dotted line: τ=0.3; dashed: τ=0.6, the pre-registered bars.

required that, at α=0, AS rise with ρ_t with Spearman ≥0.8 and that the log-log slope CI exclude 0, on all three models.

Result. The test passes on all three models (Spearman 1; slopes 0.38–0.69, all CIs far from 0; Table 6 in Appendix B). AS therefore grows with null energy that carries no image-specific content. We use *confound* in this qualified sense: AS has a non-semantic path from null energy to the score. Planted content does raise AS at fixed energy (×1.04–1.09 at α=0.1 and ×1.35–1.51 at α=0.3, ρ_t=0.2). AS is sensitive to leakage as well, but at these strengths it moves less than with a change of energy across the observed range of ρ.

Leak-free scores reproduce the order. On all 13 models, replacing each model's null content with its own leak-free surrogate, at its own energy and covariance, gives $\mathrm{AS}_{\mathrm{surr}}$ with τ=0.69 [0.39, 0.92] against SING (0.64 on S11) and Spearman 0.96 with $\mathrm{AS}_{\mathrm{SING}}$. The excess $\mathrm{AS}_{\mathrm{SING}}/\mathrm{AS}_{\mathrm{surr}}$ has τ=0.54 [0.14, 0.83], which

does not meet our pre-registered bar for any claim. Content-free null features with each model's own second moments thus reproduce most of SING's order.

4.3 Retrained translators

SING's translators were fit by its authors, and AS depends on them. We fit our own ridge translators $f \mapsto$ CLIP image embedding on half of the 10k images (5000; penalty chosen on a held-out split of that half) and evaluate on the other half. Test fidelity (mean cosine) is 0.79–0.84, against 0.81–0.87 for the released translators on the same images. With retrained translators, $\mathrm{AS}_{\mathrm{SING}}$ still agrees with SING (τ=0.85 $[0.61, 1.00]$; 0.82 on S11; τ=0.72 against released-translator AS). The order is thus not an artefact of the released translators.

The equal-energy result, however, is. With retrained translators, A1 keeps part of the agreement: τ=0.61 $[0.15, 0.94]$ on E12 and 0.75 $[0.32, 1.00]$ on S11. The pre-registered penalty grid put every model at its upper edge, so we extended the grid post hoc (10^3–10^6). Over the combined grid the chosen penalties are interior: 10^4 for VGG16, VGG19 and DenseNet, and 10^3, the value already used, for the other ten models. With these penalties A1 gives 0.39 $[-0.21, 0.86]$ on E12 and 0.53 on S11; the drop from 0.61 comes from re-fitting just the two VGG models, so this number is fragile. Forcing heavier regularisation on every model (10^6) raises it to about 0.79. How much of the order survives equal-energy removal therefore depends on how the translator is fit. Energy accounts for the agreement with the released translators. With our ridge translators, it accounts for part of it, and the remainder grows with the ridge penalty. We do not know which translator better reflects null semantics. We read this as a limit on the confound claim, not a reversal of it.

4.4 Are high-energy null directions more semantic?

If the high-variance null directions were the semantic ones, energy would partly *be* the signal. At matched d, we compare the top-variance null PCs with random null subspaces. In round 2 we used $G = \mathrm{AS}/\sqrt{\rho}$. The pre-registered round-2 label was "top PCs carry less AS per unit energy" on 2 of the 3 screened models at d=24: $G_{\mathrm{top}}/G_{\mathrm{rand}}$ was 0.60 for ResNeXt101 and 0.87 for DINOv2-L, with CIs below 1, against 1.16 for ResNet50. At d=100 the ratios were 1.01 (ResNet50), 0.85 (ResNeXt101), 0.83 (DINOv2-L) and 0.011 (BiT, not counted). The second review pointed out that $\sqrt{\rho}$ is not the fitted law. In round 3 we compare against $\rho^{\hat{a}}$, $H = (\mathrm{AS}_{\mathrm{top}}/\rho^{\hat{a}}_{\mathrm{top}})/(\mathrm{AS}_{\mathrm{rand}}/\rho^{\hat{a}}_{\mathrm{rand}})$. At $\hat{a}$=0.431 and d=24, H is 1.50 (ResNet50), 0.74 (ResNeXt101) and 0.98 $[0.97, 0.99]$ (DINOv2-L), which gives the pre-registered label "top PCs carry less AS than energy predicts" (2/3). At the low end of the CI for a (0.388), the label flips to "more", so it is *not robust* (Table 7 in Appendix B). High-energy null directions are not clearly more semantic per unit energy. The result varies across models and with the exponent, so this test does not decide between the readings.

4.5 What the interventions show

Energy and SING's score cannot be separated by their agreement with SING's order (§3). Equalising energy with SING's own translators removes the agreement. Content-free null features with each model's energy and covariance reproduce most of the order. Under retrained translators part of the agreement remains after equalisation, and the PC-band test is not decisive. These results show that AS has a non-semantic energy path large enough to produce SING's cross-model order. They do not show that AS contains no leakage signal, and the planted-concept test shows that it does respond to leakage at fixed energy.

5 The DinoViT checkpoint

SING reports its "DinoViT" at 84.0% top-1 and itself describes the model as pretrained beyond ImageNet. Its released configuration loads a timm AugReg ViT-L/16. We evaluated five public ViT-L/16 checkpoints on the full 50k validation set: torchvision IMAGENET1K_V1 (79.67%), timm AugReg (85.85%), SWAG linear (Singh et al., 2022) (85.14%), SWAG end-to-end (88.07%) and `google/vit-large-patch16-224` (81.98%) (Fig. 7, Appendix C). None falls within 84.0 ± 0.3. SING's evaluation protocol is not specified, so the 84.0% versus 85.85% gap is unexplained.

With SING's `dinovit1` translator on 1000 images, the timm AugReg features give mean cosine 0.848 to the CLIP image embeddings. Their shuffled-pair floor is 0.575, centred cosine is 0.674 and retrieval top-1 is 78.8%. Every other candidate, re-extracted with its native transform, is at its own shuffled floor (largest gap 0.002) with retrieval top-1 at most 0.2%. SING's translator was thus fit on this supervised timm checkpoint, while the other candidates are at chance. DINO never released a ViT-L with a 1000-way head (Caron et al., 2021). SING's narrative about self-supervision should therefore not rest on this model. We have not evaluated a true DINO backbone with a linear head (a GPU job specification is in the repository).

6 Size-corrected scores and SING's claims

Size-corrected scores. $\mathrm{AS}_{\mathrm{fair}} = \mathrm{AS}_{\mathrm{null}}/\mathrm{AS}_{\mathrm{rand}}$ divides AS by the AS of an energy-matched random removal. We used three families of controls, and name the family each time. In *family A*, the random subspace is Haar over the full feature space and changes the logits. *Family B* draws from the lowest-σ row-space directions of W. It is not logit-preserving either: it moves the logits by up to 14–74 logit SDs on models with a large span (VGG, ResNeXt; Table 5). *Family C* compares a 24-dimensional null subspace with a family-B control of matched energy. Point τ against SING on all 13 models is -0.18 (A), -0.41 (B) and -0.13 (C), and on S11 it is -0.56, -0.60 and -0.27. All fall below the pre-registered 0.3 bar.

Over eight normalisers (AS/ρ^a for $a \in \{0.25, 0.43, 0.5, 1\}$, $\mathrm{AS}/d_{\mathrm{null}}$, families A–C), the 95% CI excludes $\mathrm{AS}_{\mathrm{SING}}$'s 0.82 for 7 of 8 normalisers on all 13 models. On S11 it does so for exactly 5 of 8, so the weak "differs from" criterion passes at the threshold (5/8). The pre-registered reversal criterion (CI entirely below 0 under at least 5 of 8 normalisers) fails. It fails on all 13 models (0/8), and on S11 only family B meets it ($[-0.89, -0.23]$). That single pass comes from a control that is not logit-preserving. Normalising by the fitted exponent leaves a positive τ ($\rho^{0.43}$: $+0.33$ $[-0.15, 0.77]$; $\sqrt{\rho}$: $+0.26$), and family-A $\mathrm{AS}_{\mathrm{fair}}$ over-corrects: its Spearman correlation with ρ is -0.79 $[-1, -0.23]$ on S11 (Fig. 5, Appendix C). Size-corrected rankings therefore *differ* from SING's, but the direction depends on the normaliser, and we make no reversal claim. Model ranks from $\mathrm{AS}_{\mathrm{SING}}$ and $\mathrm{AS}_{\mathrm{fair}}$ also shift with the null tolerance. Adjacent-tolerance τ falls to 0.87 and 0.82, below the pre-registered 0.9, while ρ stays at or above 0.92.

Class-level claims. For each of SING's testable claims we define a signed effect (positive = SING's direction) and bootstrap images within class (1000×, BCa intervals; Table 3). A claim *holds* if the CI is above 0, is *reversed* (rev.) if below, and is *not confirmed* (n.c.) if it covers 0. Rows that share a computation are merged. Under $\mathrm{AS}_{\mathrm{SING}}$, 7 claims hold, 3 reverse and 1 is contradicted. The reversed claims are C4 (DinoViT does not have the lowest class AS; ConvNeXt and DenseNet are lower), C8 (ResNet's porcupine and sports-car classes are not in its top decile) and C10 (jellyfish ≫ camel; effect -0.107, BCa $[-0.210, -0.006]$). C9 ("no correlation" between ResNet and DinoViT class AS) is contradicted by a small correlation (-0.033, BCa CI excludes 0). These differences may reflect protocol differences (for C10, SING used about 30 concepts and 1300 images). They are not caused by size correction. Under $\mathrm{AS}_{\mathrm{fair}}$ (family A) the other testable effects fall on the opposite side (opp.). Because $\mathrm{AS}_{\mathrm{fair}}$ over-corrects, we treat these as diagnostics, not reversals. The image-bootstrap CIs ignore uncertainty from the choice of models and normaliser, so they are narrow. One BCa interval (the $\mathrm{AS}_{\mathrm{fair}}$ class-SD effect of C6) does not contain its point estimate, because the bootstrap distribution is nearly degenerate and the BCa adjustment is unstable. We flag it and do not use it.

ID	Claim (short)	Effect AS_{SING} [BCa 95% CI]	AS_{SING}	AS_{fair}
C1/C11	DinoViT best ratio / least steerable (proxy)	$0.149\ [0.142, 0.156]$	holds	opp.
C2	ResNeXt worst ratio (proxy)	$0.323\ [0.303, 0.343]$	holds	opp.
C3/C6m	ResNet > DinoViT (class mean)	$1.383\ [1.372, 1.394]$	holds	opp.
C4	DinoViT least leakage	$-0.293\ [-0.298, -0.289]$	**rev.**	opp.*
C5	ResNeXt top-2 of 5	$1.442\ [1.424, 1.462]$	holds	opp.
C6 sd	ResNet class SD > DinoViT	$0.744\ [0.735, 0.759]$	holds	opp.§
C7	DinoViT per-class $\lvert AS\rvert < 1°$	$0.500\ [0.499, 0.500]$	holds	n/a
C8 porcupine	Porcupine ≥90th pct (ResNet)	$-5.2\ [-8.4, -1.9]$	**rev.**	opp.
C8 sports car	Sports car ≥90th pct (ResNet)	$-29.0\ [-34.1, -23.9]$	**rev.**	opp.
C9 spearman	No ResNet–DinoViT class corr.	$-0.033\ [-0.074, -0.012]$	contra.	contra.
C10	Jellyfish ≫ camel	$-0.107\ [-0.210, -0.006]$	**rev.**	opp.
C12	Supp. Fig. 12b order (τ)	$0.821\ [0.56, 1.00]^{\ddagger}$	holds	n.c.‡
C13	EVA02 target-space order	–	untestable	untestable

Table 3: SING's class-level claims under AS_{SING} (BCa 95% CI, images resampled within class, so the CIs are conditional on the 13 fitted translators) and family-A AS_{fair}. rev. = CI on the opposite side of 0; n.c. = not confirmed; opp. = opposite side under AS_{fair}, a diagnostic since AS_{fair} over-corrects; contra. = a nonzero correlation where SING reports none. C1/C11 and C3/C6 mean are one computation each. C1/C2/C11 use $|AS|$ proxies for SING's ratio and steerability. *Under AS_{fair}, C4 coincides numerically with C1/C11 (same model attains the minimum) and is counted once. §BCa interval excludes its point estimate (degenerate bootstrap); not used. ‡Model bootstrap; under AS_{fair}, $\tau = -0.18\ [-0.78, 0.35]$, not confirmed. C7 has no AS_{fair} analogue.

7 Discussion

What we can and cannot conclude. SING's cross-model ranking is indistinguishable from the ranking by null energy. With SING's released translators, removing equal energy removes the agreement, and leak-free null content at each model's own energy reproduces most of it. AS thus responds to null energy through a path that carries no image-specific content. We do not conclude that SING's ranking is wrong, or that AS carries no semantic signal. AS responds to planted concepts at fixed energy, and with our retrained translators part of the order survives equal-energy removal. The claim we support is narrower: a cross-model AS comparison cannot be read as a comparison of null semantics unless energy is controlled, and the result of that control depends on the translator.

Why does ρ vary? Neural collapse (Papyan et al., 2020) offers a candidate mechanism. Late in training, features concentrate near class means aligned with the classifier rows, so the energy outside the row space shrinks. Models trained further into that regime, or with heads fit differently, would have smaller ρ and hence smaller AS. We have not tested this.

Relation to other null-space work. ActSub (Zöngür et al., 2025) finds that the activation components a classifier largely ignores help to detect distribution shift, and Jacobsen et al. (2019) show that class information can hide in directions the logits do not use. Both are consistent with null content being informative. Cross-model scores of that content, however, need an energy-only baseline. Linear erasure methods (Ravfogel et al., 2020; Belrose et al., 2023) remove a concept so that no linear probe can recover it, a stricter notion than projecting out $\ker W$. A leakage score built on them would ask whether class information is linearly recoverable from the null component rather than how far the translation moves. Linear translators into CLIP (Moayeri et al., 2023) differ in fidelity across models, and §4.3 shows that the translator affects interventional conclusions.

Recommendations.

1. Report d_{null}, ρ and the null tolerance next to any null-space score, and check rank stability across tolerances.

2. Compare every cross-model score with the ranking by ρ alone, as a control in the sense of Hewitt & Liang (2019), and with an equal-energy intervention.
3. Check conclusions with a second, independently fit translator.
4. Name the exact checkpoint and its measured accuracy, and check translator fidelity against a shuffled floor.
5. Flag translator conditioning when full-null and partial-null AS disagree (BiT).
6. Validate flatness or invariance criteria on a synthetic oracle before using them to accept or reject scores (Appendix D).

Limitations. There are only 13 models, so every cross-model CI is wide. SING's order is digitised from a figure. Our retrained translators are ridge maps chosen by us. SING's training recipe may differ, and the equal-energy result depends on this choice. The leak-free test covers three models with d_{null}=1048. We did not calibrate the energy exponent a on a synthetic model with a known translator. The energy law does not cleanly separate energy from dimension, and its pre-registered flatness part fails. The class-claim CIs reflect image resampling only, several claims rely on proxies, and C13 is untestable. We did not evaluate a true DINO backbone, nor information-theoretic leakage measures for concept models (Makonnen et al., 2025), which target a different notion of leakage.

8 Conclusion

Using SING's code and released translators, we reproduce its 13-model ranking, and the null energy fraction ρ reproduces it equally well. With the released translators, equalising null energy removes the agreement, and image-independent null content with each model's own energy and covariance reproduces most of the order. In that qualified sense, energy is a confound of the cross-model comparison. With retrained translators, part of the agreement survives equalisation, so the strength of this conclusion depends on the translator. SING's "DinoViT" translator matches a supervised timm checkpoint. Size-corrected scores change the order without reversing it with confidence. Cross-model null-space comparisons should report ρ, an energy-only baseline and an equal-energy intervention next to the score.

Reproducibility. The supplementary material contains the code, pre-registrations, aggregated results and a one-command `make reproduce` that regenerates every table, figure and number in this paper from committed aggregates, and the repository will be made public on acceptance. ImageNet-derived features are not redistributed; checksums, the image list and the extraction scripts are included. A full rerun from features takes about 2 h on 8 CPU cores; feature extraction needs a GPU and ImageNet (Appendix E).

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A Pre-registration ledger

Table 4 lists every pre-registered experiment and its outcome. E1 to E9 come from the original plan, R2 from the round-2 pre-registration and R3 from the round-3 one, each written in response to reviews.

Exp.	Pre-declared criterion (short)	Outcome
Exp 0	null size correlates with SING IS/AS	GO rests on ρ (Spearman -0.94); d_{null} alone -0.46, p=0.12
Exp 1	r=AS(1048)/AS(24) ≥ 3 on $\geq 3/4$; $\tau_{d=24} \leq 0.5$	GO ($r \approx 10$ on 3/4; BiT 0.06; τ=0.29)
E1	high/low $\geq 3\times$ at fixed d; $\leq 1.5\times$ across d at fixed ρ	MIXED (first part 4/4; second fails)
E2	$\tau(\text{AS}_{\text{fair}}, \text{SING}) \leq 0.5$ or headline claim reverses	PASS as registered; weakened in round 2
E3	$\tau(\text{AS}_{\text{fair}}, \text{planted}) \geq 0.8$, $\tau(\text{AS}_{\text{SING}}, \text{planted}) \leq 0.5$	not passed (0.109 vs 0.386); KILL not met
E8	NCP probe $\tau \geq 0.8$ and flat; KILL if $\tau < 0.6$	KILL (τ=-0.331; flat 0/4)
E9	20-candidate screen, DEV $\tau \geq 0.8$ and flat on $\geq 3/4$	KILL (best M1 τ=0.976, flat 0/4)
R2	KILL-R: drop "disagrees" unless fix 2 and D-weak pass	not fired; "differs from" allowed; D-weak screened at the threshold (5/8)
R2	D-strong: τ CI < 0 under $\geq 5/8$ normalisers	FAIL (0/8 all 13; 1/8 screened)
R2	AS_{fair} removes the ρ trend	FAIL on screened set (-0.79 [$-1.00, -0.23$])
R2	tolerance stability: adjacent $\tau \geq 0.9$	FAIL (AS_{fair} 0.87/0.82; AS_{SING} 0.87); $\rho \geq 0.92$
R2	PC bands $G_{\text{top}}/G_{\text{rand}}$ vs 1 at d=24	"top PCs carry less AS per unit energy" (2/3 screened models)
R2	synthetic oracle passes the flatness rule	FAIL (ratio 3.55): rule ill-posed
R2	DinoViT translator fidelity: used checkpoint best by ≥ 0.05	PASS (0.848 vs 0.677); see D
R3-A	equal-energy A1: ENERGY-ACCOUNTS / PARTIAL / KILL-R3	ENERGY-ACCOUNTS ($\tau = -0.03$ E12, 0.09 S11); no KILL
R3-A	A4 (matched CLIP energy) $\tau < 0.3$ on both sets	yes (-0.15, -0.20); $\kappa - \rho$ CI covers 0
R3-B	N4: AS rises with leak-free null energy, all 3 models	PASS (Spearman 1; slopes 0.38–0.69)
R3-B	B-rank: $\tau(\text{AS}_{\text{surr}}) \geq 0.6$ on all 13 and S11	yes (0.69, 0.64); no claim from X
R3-C	PC bands vs energy law $\rho^{\hat{a}}$	"top PCs carry less AS than energy predicts"; not robust in $\hat{a}$
R3-D	N3: non-matching candidates at chance	PASS ($\lvert\cos - \text{shuffled}\rvert \leq 0.002$)
R3-E	N7: no claim CI excludes its point	FAIL (1 flag: AS_{fair} C6 SD); N6 dedup done
R3-F	N1: paired $\tau(\rho) - \tau(\text{AS}_{\text{SING}})$ CI covers 0	yes [$-0.36, 0.29$]: "indistinguishable"
R3-G	retrained translators: order survives if $\tau \geq 0.6$ on all 13 and S11	survives (0.85, 0.82); A1 with retrained translators 0.61, 0.75 (descriptive)

Table 4: All pre-registered experiments and their outcomes. The pre-registration files are in the repository.

B Additional tables

Model	d_{null}	ρ	κ	γ	AS_{SING}	A1	AS_{surr}	fair A	fair B	B max Δlogit/SD
VGG16	3096	0.350	0.119	0.34	1.088	0.084	0.922	0.708	0.867	14.4
VGG19	3096	0.339	0.123	0.36	1.256	0.094	0.879	0.716	1.087	13.7
DenseNet121*	27	1.8×10^{-8}	2.5×10^{-8}	1.36	2.5×10^{-4}	–	2.4×10^{-4}	0.152	0.671	1.5×10^{-8}
ResNet50	1048	0.146	0.281	1.93	1.676	0.160	1.176	0.855	0.989	0.3
"DinoViT"	24	0.005	0.013	2.46	0.294	0.284	0.296	1.632	1.220	5.0×10^{-3}
EffNet-B0	280	0.130	0.030	0.23	0.443	0.076	0.378	0.889	1.267	0.8
BiT-R50x1*	1048	0.046	0.061	1.31	0.557	0.153	0.589	0.042	1.142	9.3×10^{-2}
ResNeXt101	1048	0.451	0.219	0.49	1.999	0.112	1.476	0.707	0.591	74.4
ConvNeXt-B	24	0.013	0.002	0.17	0.080	0.049	0.085	0.974	0.961	4.1×10^{-2}
Swin-L	536	0.274	0.151	0.55	1.019	0.105	1.170	0.829	1.130	7.5
DINOv2-L	1048	0.322	0.165	0.51	1.499	0.104	0.960	0.712	0.961	6.0
DeiT3-L	24	0.348	1.085	3.11	5.048	0.215	3.364	1.065	0.776	16.4
EVA02-L	24	0.009	0.031	3.47	0.417	0.302	0.411	2.612	1.306	2.3×10^{-2}

Table 5: Per-model scores on the 10k subset at tolerance 10^{-5}. κ: CLIP-space null energy; $\gamma = \kappa/\rho$; A1: AS after removing $\sqrt{\rho_0/\rho}\,\Pi f$; AS_{surr}: AS with leak-free surrogate null content; fair A/B: AS_{fair} families; last column: largest logit change of the family-B control in logit SDs (family B is not logit-preserving). *Excluded by the screen. Family C: Table 8.

	AS at α=0, ρ_t =					AS(α)/AS(0), ρ_t=0.2	
Model	0.01	0.05	0.2	0.45	slope [95% CI]	α=0.1	α=0.3
ResNet50	0.371	0.665	0.817	1.935	0.377 [0.375, 0.379]	1.088	1.508
ResNeXt101	0.074	0.175	0.453	1.105	0.690 [0.689, 0.692]	1.040	1.353
DINOv2-L	0.082	0.192	0.485	1.151	0.677 [0.676, 0.678]	1.049	1.363

Table 6: Leak-free null content (§4.2). AS at α=0 against the surrogate's null energy ρ_t; OLS slope of logAS on $\log \rho_t$ with 1000× image bootstrap; AS response to a planted concept at ρ_t=0.2.

	d=24, $\hat{a}$ =			d=100, $\hat{a}$ =		
Model	0.431	0.388	0.485	0.431	0.388	0.485
ResNet50	1.50 [1.48, 1.51]	1.75 [1.73, 1.77]	1.23 [1.21, 1.24]	1.18 [1.17, 1.19]	1.30 [1.28, 1.31]	1.05 [1.03, 1.06]
ResNeXt101	0.74 [0.73, 0.76]	0.85 [0.83, 0.87]	0.63 [0.61, 0.64]	0.97 [0.96, 0.99]	1.06 [1.04, 1.07]	0.87 [0.86, 0.89]
DINOv2-L	0.98 [0.97, 0.99]	1.06 [1.04, 1.07]	0.89 [0.88, 0.90]	0.91 [0.89, 0.92]	0.96 [0.95, 0.97]	0.85 [0.83, 0.86]
BiT-R50x1*	0.009 [0.009, 0.009]	0.010 [0.010, 0.010]	0.008 [0.008, 0.008]	0.012 [0.012, 0.012]	0.013 [0.013, 0.013]	0.011 [0.011, 0.011]

Table 7: PC bands against the fitted energy law (§4.4): H with 1000× image-bootstrap CIs at three values of $\hat{a}$ (E1 point estimate and CI ends). $H < 1$: top PCs carry less AS than energy predicts. *BiT is reported but not counted.

	VGG16	VGG19	Dense	RN50	DinoViT	Eff	BiT	RNX	ConvNeXt	Swin	DINOv2	DeiT3	EVA02
fair C	1.32	1.35	0.71	1.14	1.20	1.44	135.6	1.04	0.97	1.08	1.00	0.79	1.33

Table 8: Family-C AS_{fair} per model (`r2/results/fix2_models.csv`).

C Energy law, Exp 1 and checkpoint accuracy

Model	r	2σ band	R^2 (vs $d \equiv$ vs ρ_d)
ResNet50	10.15	[5.83, 14.48]	0.983
ResNeXt101	11.44	[6.93, 15.95]	0.977
DINOv2-L	10.48	[8.85, 12.10]	0.974
BiT-R50x1	0.060	[0.057, 0.063]	0.512

Table 9: Exp 1: $r = \mathrm{AS}(d{=}1048)/\mathrm{AS}(d{=}24)$ over random null subspaces (20 seeds). For random subspaces ρ_d is linear in d, so the two R^2 coincide. At matched $d{=}24$ across 8 models, τ against SING is 0.29 ($p{=}0.40$).

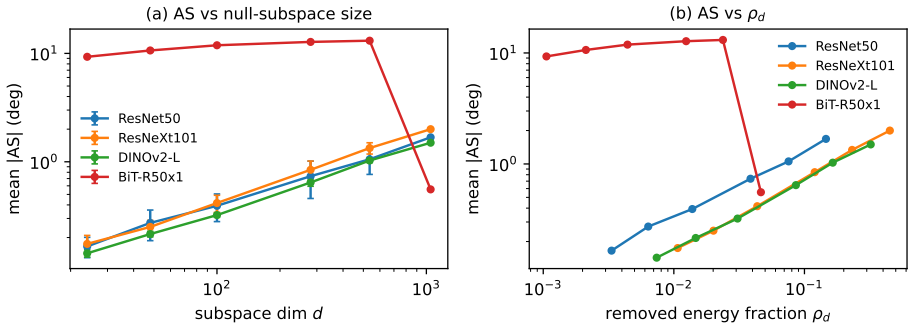


Figure 3: Exp 1: mean |AS| against subspace size d (a) and removed energy ρ_d (b). BiT collapses at the full null (translator cancellation). Full-null anchors: ResNet50 1.682°, ResNeXt101 1.999°, BiT 0.556°, DINOv2-L 1.499°.

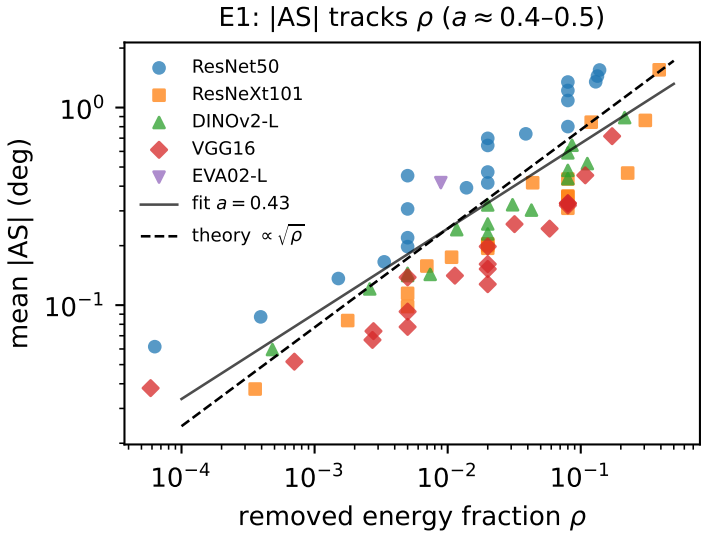


Figure 4: Energy law: |AS| against ρ over 76 cells on five models (ResNet50, ResNeXt101, DINOv2-L, VGG16, EVA02-L) at fixed $d \in \{24, 100, 280\}$ and fixed $\rho \in \{0.5\%, 2\%, 8\%\}$. Solid: pooled fit ($a$=0.431); dashed: $\sqrt{\rho}$. At fixed d the high/low-energy ratio exceeds 3× on all 4 evaluable models; at fixed ρ the max/min across d exceeds 1.5 in several cells (up to 2.29), so the pre-registered rating is *mixed*.

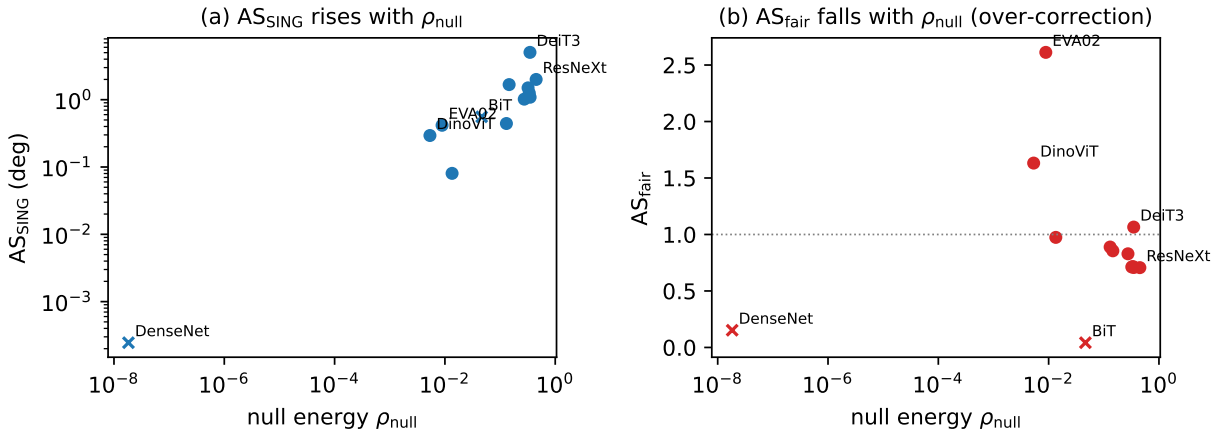


Figure 5: (a) AS_{SING} rises with ρ. (b) Family-A AS_{fair} falls with ρ: it over-corrects. $\times$: models excluded by the screen.

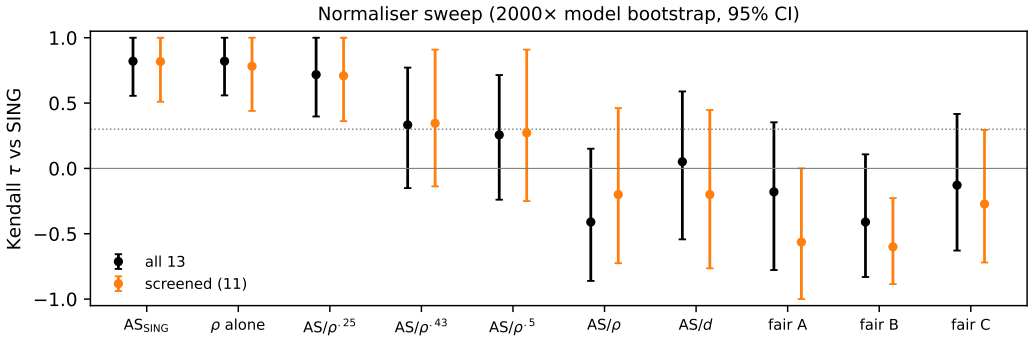


Figure 6: τ against SING for each normaliser and for ρ alone (2000× model bootstrap), on all 13 models and on S11. Dotted line: the τ=0.3 bar.

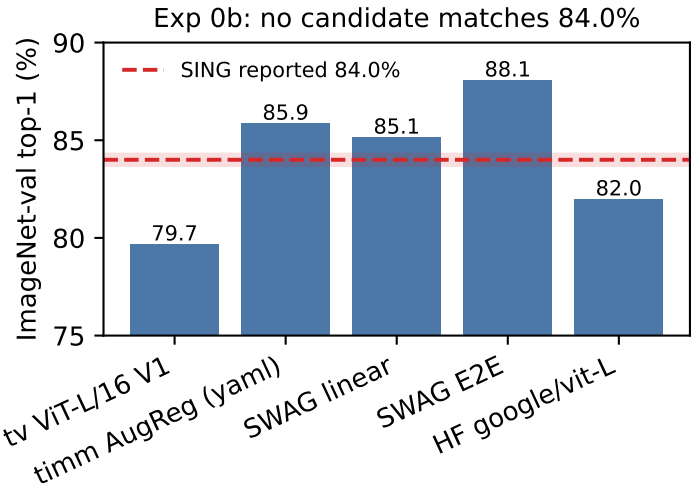


Figure 7: Top-1 on ImageNet-val (native evaluation transform) of the five public ViT-L/16 candidates. None matches SING's reported 84.0%.

D Planted leakage, score search and the flatness rule

Planted leakage (E3). We planted a CLIP concept direction (`red`, `striped`, `water`) into the null spaces of 8 models at log-uniform $\alpha \in [0.01, 1]$ over 50 trials, with logits unchanged (max $|\Delta\text{logit}| \leq 2.3\times10^{-5}$). Kendall τ against the planted ranking, averaged over concepts, is 0.386 $[0.324, 0.448]$ for AS_{SING} and 0.109 $[0.046, 0.171]$ for AS_{fair} (both α=0-subtracted). The pre-registered target ($\text{AS}_{\text{fair}} \geq 0.8$, $\text{AS}_{\text{SING}} \leq 0.5$) was not met. Neither score recovers planted strength well across models.

E8 (pre-registered; killed). A probe-based null-content score (ridge-probe R^2 on null coordinates, z-scored against energy-matched random subspaces) gave $\tau{=}{-}0.331$ $[-0.392, -0.273]$ on the planted task and was flat on 0/4 models. Three implementation caveats apply: the z-score saturates, λ was chosen by approximate leave-one-out rather than the pre-registered exact cross-validation, and λ sat at the grid floor in 18 of 96 fits.

E9 (pre-registered; killed). We screened 20 candidate scores on two development concepts. The rule required $\tau \geq 0.8$ and flatness (max / min ≤ 1.5 across nested null subspaces) on at least 3 of 4 models at $\alpha \in \{0.1, 1\}$. No candidate passed. The best was M1, a debiased null-leakage estimate $\hat{L}^2 = (\|\sum_i Pf_i s_i\|^2 - \sum_i \|Pf_i\|^2 s_i^2)/(n(n{-}1)\mathbb{E}\|f\|^2)$ with standardised concept scores s_i, scored as $\sqrt{\max(\hat{L}^2, 0)}$. It reached τ=0.976 $[0.963, 0.988]$ but was flat on 0/4 models.

The flatness rule is ill-posed. We tested the rule on synthetic Gaussian features with a known head (m=400, d_{null}=300). An exact oracle, the planted amplitude inside the removed subspace, fails it with max / min 3.55 and 3.48 ($\approx \sqrt{300/24}$), because a larger removed subspace contains more of the plant. A revised rule (max / min of score/oracle ≤ 1.5, post hoc) is passed by M1 (ratios 1.00–1.03). Under the revised rule M1 is flat on 4/4 models, and E8's score on 0/4. E8 and E9 remain killed as registered.

Holdout (post hoc). We ran the E9 holdout on four held-out concepts (water, green, thin, knitted; 50 trials at seed 1; linear and nonlinear plants) for the six candidates declared in the round-2 pre-registration plus AS_{SING} and AS_{fair}, with the E9 pass rule (Table 10). Only M1 meets both τ thresholds. Under the registered flatness rule no candidate passes; that rule also fails for the oracle (above). Under the revised rule, M1 passes. Paired over trials (1000×, seed 0), $\tau(\text{M1}){-}\tau(\text{AS}_{\text{fair}})$ is 0.77 [0.71, 0.82] for the linear plant and 1.01 [0.96, 1.06] for the nonlinear plant, so M1 tracks the planted leakage better than AS_{fair} under both

Score	lin. τ [95% CI]	nl. τ	flat	rev. flat	pass
M1	0.93 [0.91, 0.95]	0.97	0/0	4/4	rev.
M2	0.44 [0.39, 0.48]	0.46	0/0	0/0	no
S9	0.45 [0.40, 0.51]	0.44	0/1	2/0	no
S10	0.37 [0.33, 0.42]	0.52	3/4	0/0	no
M5 ($AS/\sqrt{\rho}$)	0.22 [0.18, 0.27]	0.15	0/0	0/0	no
S4 (AS/ρ)	−0.08 [−0.15, −0.01]	−0.14	0/0	0/0	no
AS_{SING} (raw)	0.33 [0.27, 0.40]	0.36	0/0	0/0	no
AS_{fair} (sub.)	0.17 [0.11, 0.22]	−0.04	0/0	0/0	no

Table 10: E9 holdout (post hoc; AS_{fair} is α=0-subtracted; water, green, thin, knitted; 50 trials, seed 1; 8 models). Flat = number of the 4 large-null models flat at α=0.1/1 under the registered rule; rev. flat = under the revised score/oracle rule. Pass: lin. $\tau \geq 0.8$, CI lower ≥ 0.7, nl. $\tau \geq 0.6$, flat on $\geq 3/4$ at both α (yes = registered rule, rev. = only with the revised rule).

plants. The revised rule also changes which candidate counts as flat: S10, the only candidate flat under the registered rule (3/4 and 4/4 models at α=0.1/1), is flat on 0/4 and 0/4 under the revised one. The plant is written along the null projection that M1 itself measures, and M1, unlike the AS-family scores, is given the concept score that defines the plant. This comparison therefore favours M1 by construction. One of the four held-out concepts, water, was also an E3 plant concept whose results we had seen before choosing the holdout set; its inclusion was pre-registered, but the set is not fully unseen. M1 was picked after the DEV results and the flatness rule was revised after the oracle test, so this holdout is post hoc. E9 stays a KILL as registered, and M1 is a candidate for a future pre-registered test, not a validated score.A reduced-draw mode (5 instead of 20 random draws for AS_{fair}, 2 instead of 5 nested bases, S9 omitted), declared before it was run, gives the same M1 and AS_{SING} values and the same flatness verdicts. AS_{fair} moves by +0.02 (linear) and +0.06 (nonlinear), and the run takes 36 min on 8 CPU cores; both miss the targets we set ($\pm$0.03, under 30 min; Appendix E).

M1 on the 13 models (post hoc, descriptive). At α=0 on the full null, averaged over SING's 173 concepts, M1 gives τ=0.667 $[0.29, 0.94]$ against SING, an interval that includes AS_{SING}'s value. M1 orders the models much as ρ does (τ=0.79), as expected for a score scaled by total feature energy. Its holdout validation favours it by construction (the plant is written along the projection it measures), so this agreement is not independent evidence about energy. We treat M1 as a candidate for a future pre-registered test, not as a validated score.

E Reproducibility and compute

`make reproduce` regenerates every table, figure and number in this paper from committed aggregates in under a minute. This checks consistency, not the feature-level computation. `make reproduce-full` reruns the feature-level analyses from the private ImageNet features: about 2 h wall on 8 CPU cores, most of it the holdout. `make holdout-fast` runs the reduced holdout (36 min on 8 cores). Feature extraction (13 models on the 10k subset, the Exp 1 features and the five ViT-L/16 candidates on the 50k validation set) needs ImageNet and a GPU. Two jobs were timed: the Exp 1 extraction (1.4 GPU-hours on one Kaggle GPU) and the re-extraction of the five ViT-L/16 candidates for the translator check (1.9 h on two T4 GPUs, about 3.8 GPU-hours including weight downloads). The other extraction jobs were not timed; we estimate under 10 GPU-hours for all extraction.

Seeds. Family A random draws 10000+r ($r < 20$); family B 50000+r; family C subspaces 30000+j and controls 60000+j; Exp 1 20 seeds per d; E3 plants seed 0, bootstrap 10000× seed 1; model bootstrap 2000× seed 0; claim bootstrap 1000× seed 0; E9 and holdout α seed 1; flatness bases 20000+j; PC-band draws 40000+j; A3 rotations 80000+j; leak-free surrogates 70000+r; translator split permutation seed 0.